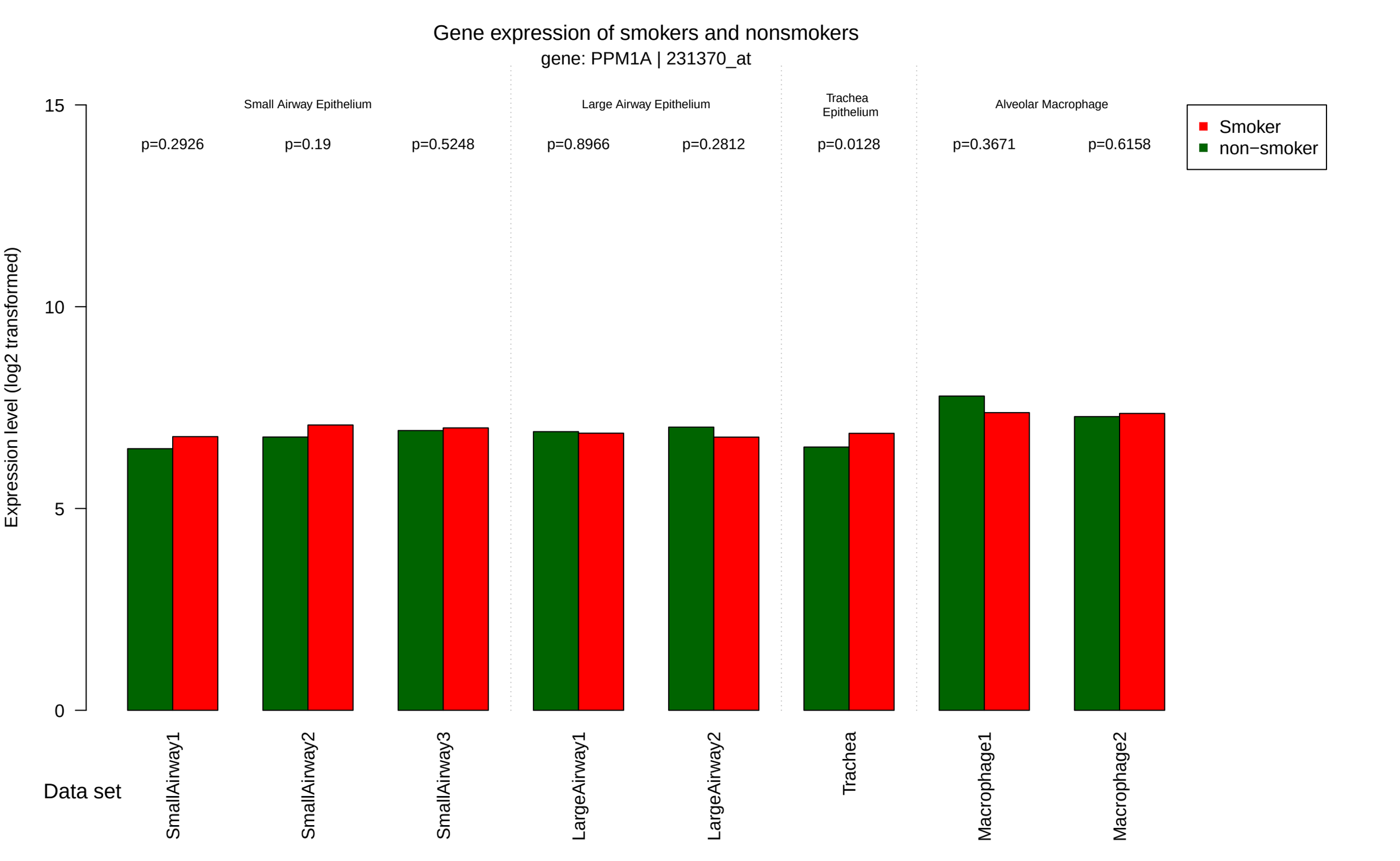
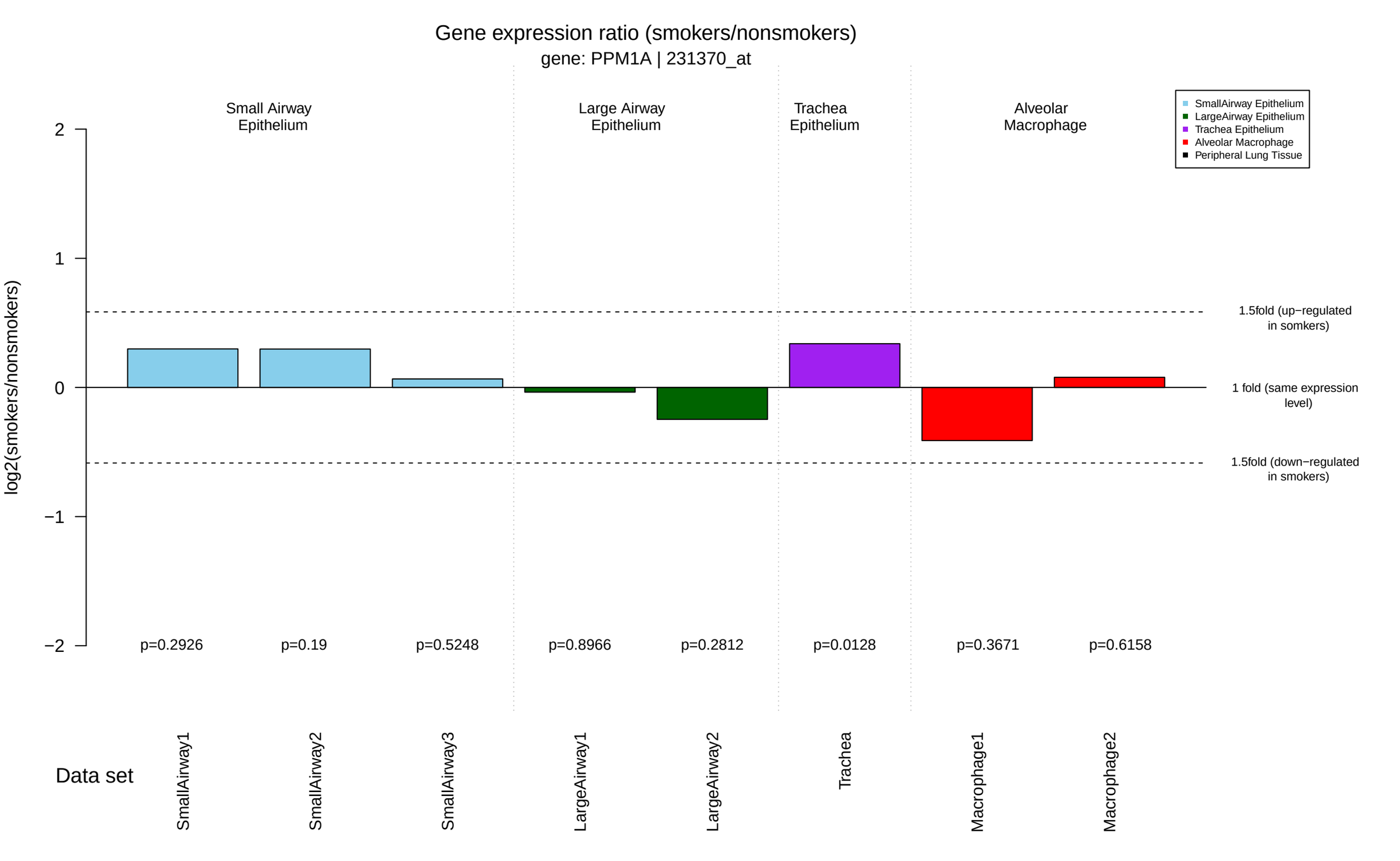
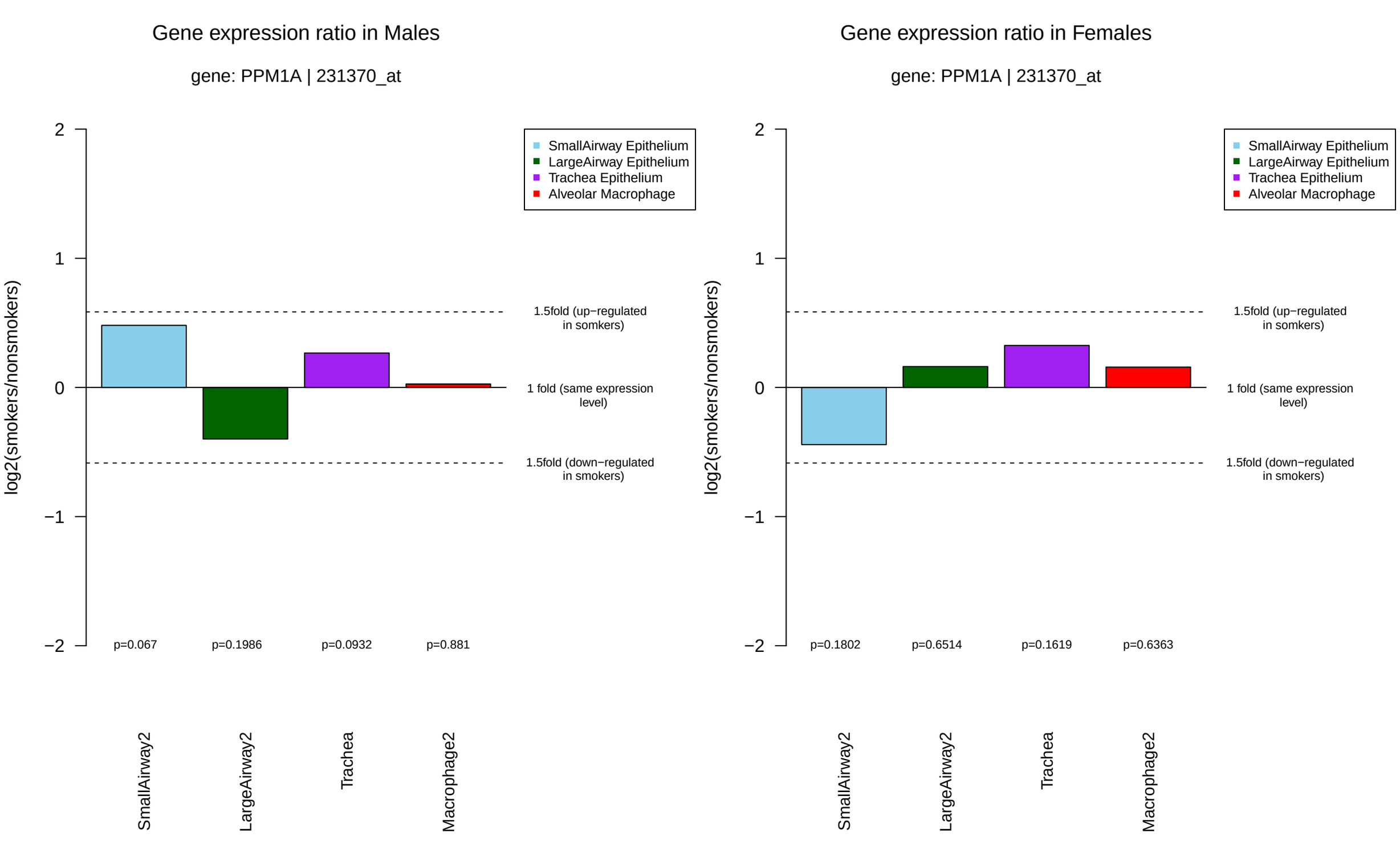
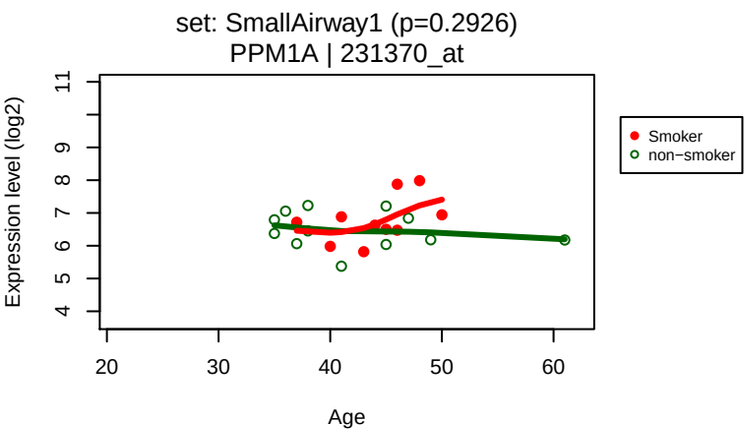
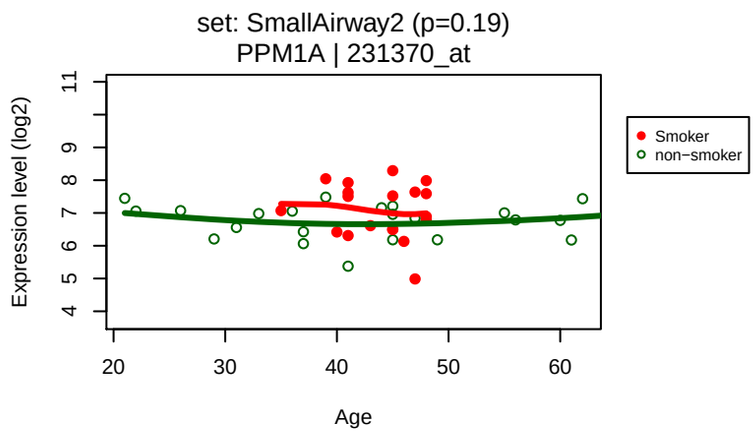
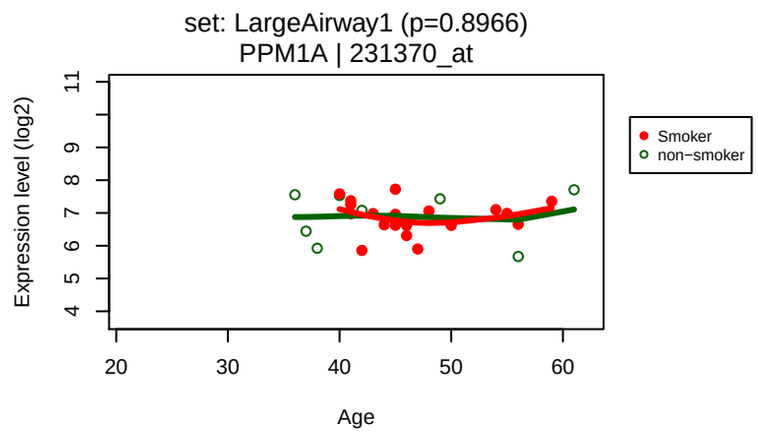

Gene expression of smokers and nonsmokers
gene: PPM1A | 231370_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2926
p=0.19
p=0.5248
p=0.8966
p=0.2812
p=0.0128
p=0.3671
p=0.6158
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

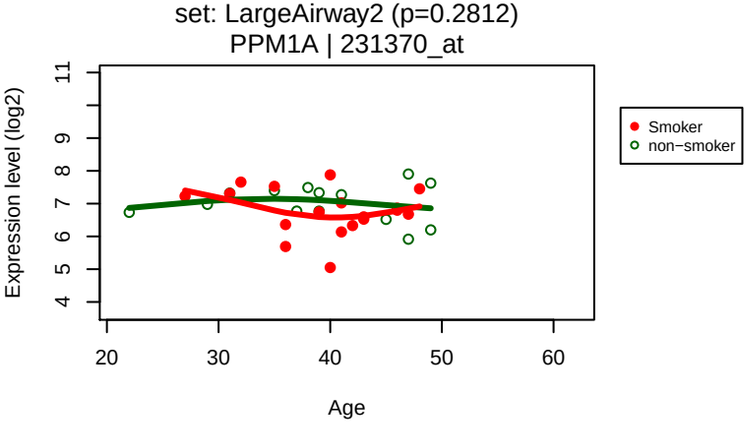

Gene expression ratio (smokers/nonsmokers)
gene: PPM1A | 231370_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2926
p=0.19
p=0.5248
p=0.8966
p=0.2812
p=0.0128
p=0.3671
p=0.6158
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

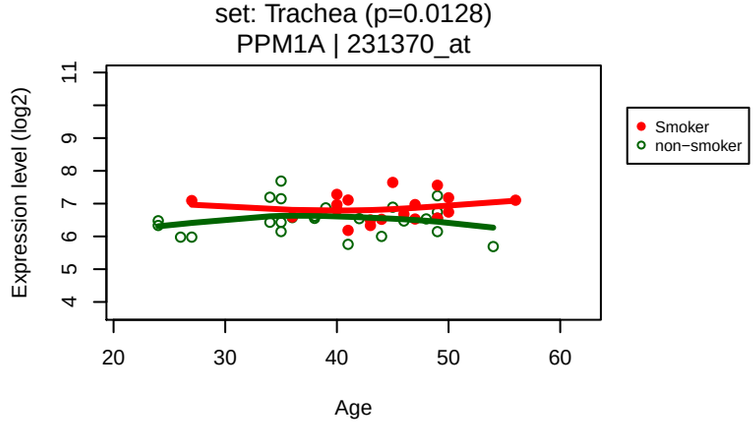

Gene expression ratio in Males
gene: PPM1A | 231370_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.067
p=0.1986
p=0.0932
p=0.881
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PPM1A | 231370_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1802
p=0.6514
p=0.1619
p=0.6363
SmallAirway2
LargeAirway2
Trachea
Macrophage2

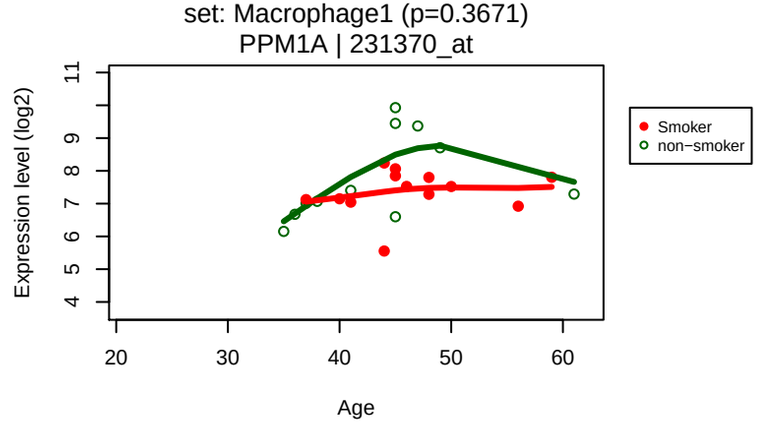

set: SmallAirway1 (p=0.2926)
PPM1A | 231370_at
Expression level (log2)
Age
Smoker
non−smoker

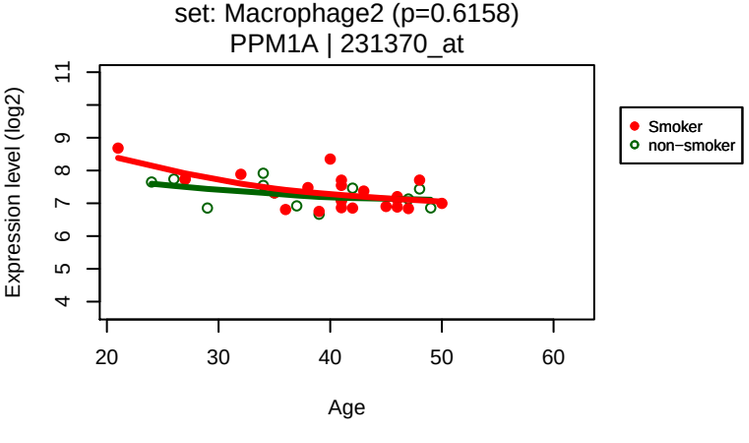

set: SmallAirway2 (p=0.19)
PPM1A | 231370_at
Expression level (log2)
Age
Smoker
non−smoker

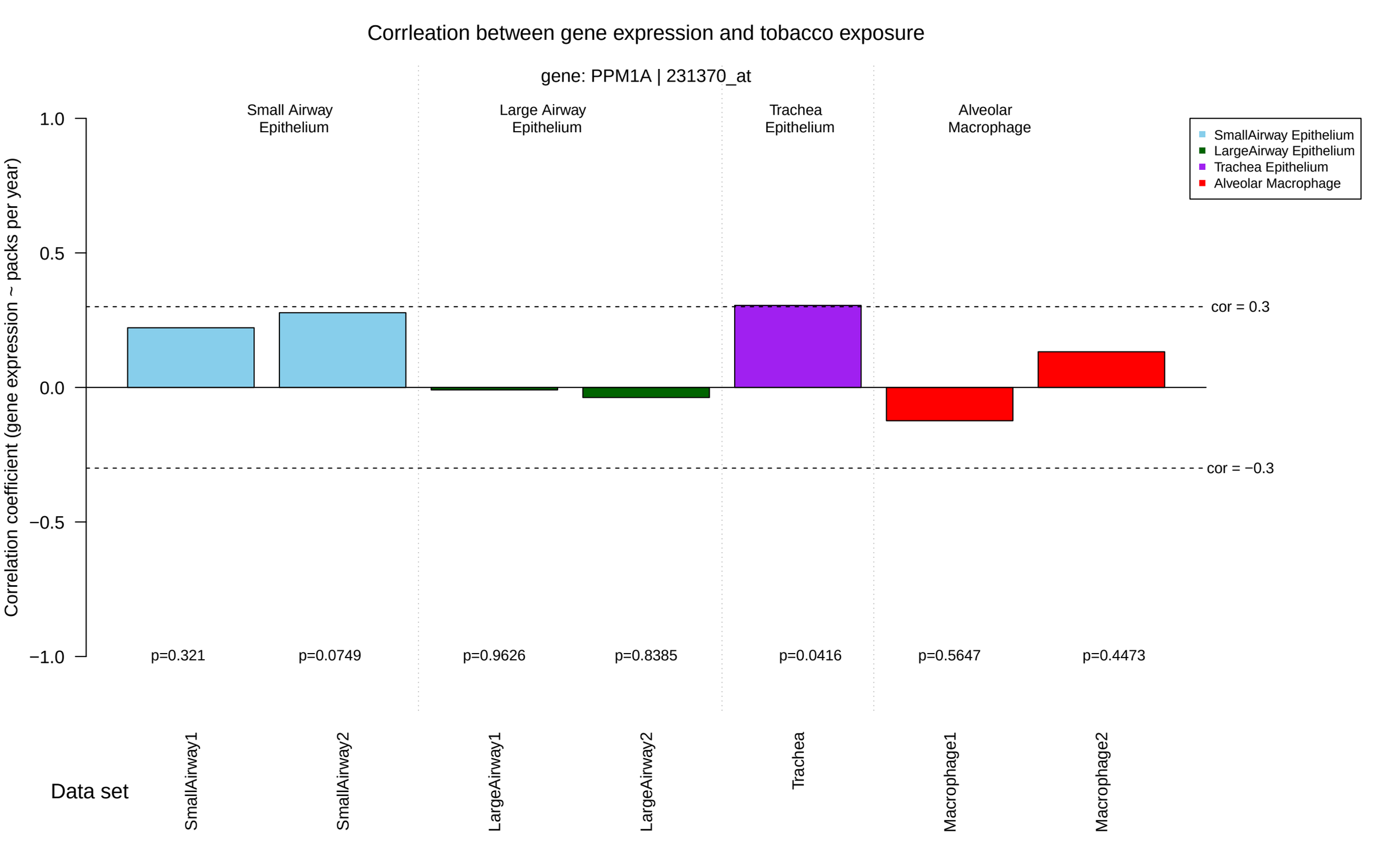

set: LargeAirway1 (p=0.8966)
PPM1A | 231370_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2812)
PPM1A | 231370_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0128)
PPM1A | 231370_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3671)
PPM1A | 231370_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6158)
PPM1A | 231370_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PPM1A | 231370_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.321
p=0.0749
p=0.9626
p=0.8385
p=0.0416
p=0.5647
p=0.4473
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2